

Session Objectives

- 1. Discuss and describe **where hematopoiesis occurs in the fetus, infant, and adult; compare red v. yellow bone marrow; define and evaluate bone marrow cellularity; discuss the monophyletic theory of hematopoiesis; and discuss extramedullary hematopoiesis.**
- 2. Discuss and describe the **concepts of CFU (colony forming units) and the growth factors which stimulate them including the different CSFs (colony stimulating factor), IL (interleukin), erythropoietin, and thrombopoietin.**
- 3. Discuss and describe the **changing morphology and evolution of the myelocyte, monocyte, and lymphocyte beginning with the blast stage and resulting in the mature white blood cell- clinical case examples.**
- 4. Discuss and describe the **changing morphology and evolution of the erythrocyte beginning with the blast stage and resulting in the mature red blood cell – clinical case examples.**
- 5. Discuss and describe the **changing morphology and evolution of the thrombocyte beginning with the blast stage and resulting in the mature platelet – clinical case examples.**

Outline

- Where hematopoiesis occurs
- Normal bone marrow and cellularity
- Extramedullary Hematopoiesis
- **Hierarchy of hematopoiesis**
 - Myelopoiesis
 - Erythropoiesis
 - Thrombopoiesis
 - Lymphopoiesis not discussed in detail, refer to immunology lectures
- Practice questions/case examples

Hematopoiesis:

- What if Bone Marrow does not function?
- **Extramedullary hematopoiesis:**
 - Occurs when bone marrow cannot meet the demands for blood cell production
 - **Liver, spleen and lymph nodes** may resume their embryonic hematopoietic activity
 - Observed in **hemoglobinopathies, myeloproliferative disorders, neoplasms involving the bone marrow, and other conditions**

Hematopoiesis: How

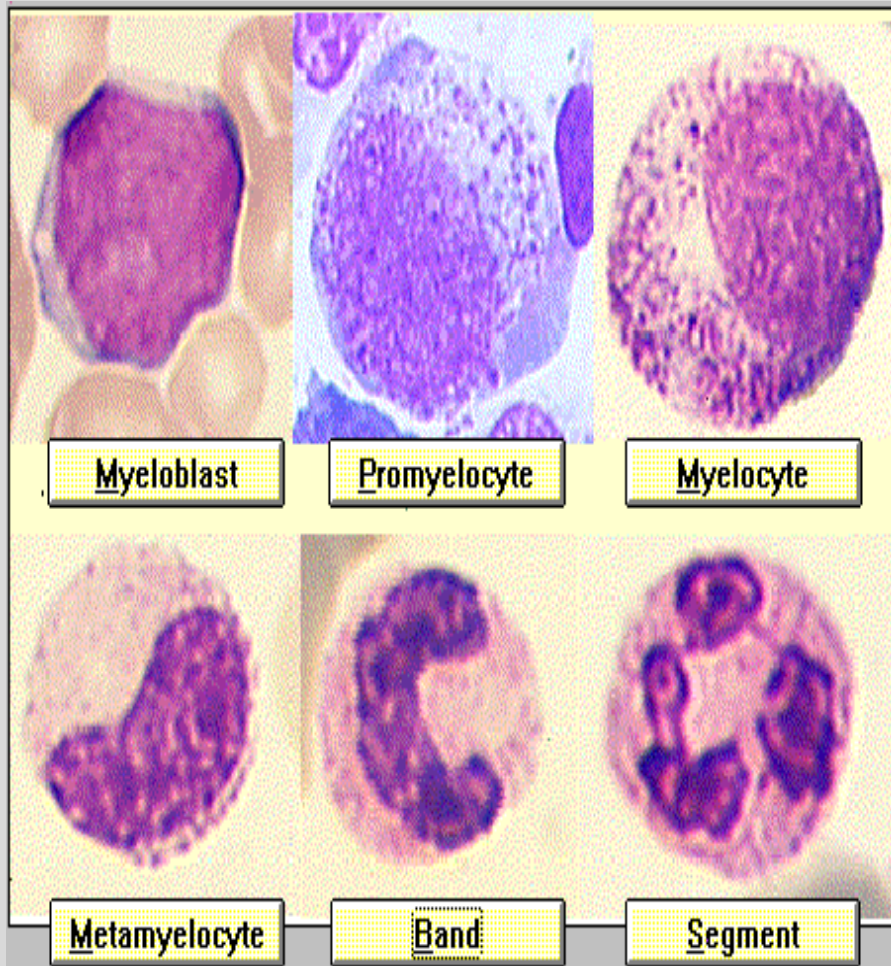
- Progenitor stem cells = Colony Forming Units (CFU):
 - Myeloid lineage: (granulocytes, monocytes, erythrocytes, megakaryocytes)
CFU – GEMM differentiates into CFU-GM, CFU-E, CFU-Meg, CFU-Eo, CFU-B
 - Lymphoid lineage: (T-cells, B-cells) **CFU-L**

Lineage	Type of CFU	Peripheral Blood cell produced
Myeloid	Erythroid-CFU (CFU-E)	Erythrocytes
Myeloid	Thrombocytic-CFU (CFU-Meg)	Megakaryocytes
Myeloid	Granulocyte-Monocyte CFU (CFU-GM)	Granulocytes, Monocytes
Myeloid	Granulocyte-Eosinophil CFU (CFU-Eo)	Eosinophils
Myeloid	Granulocyte-Basophil CFU (CFU-B)	Basophils
Lymphoid	Lymphoid lineage CFU (CFU-L)	Lymphocytes

Hematopoiesis: Growth Factors

- **What stimulates the Hematopoietic stem cell to differentiate into the Progenitor cells...and then these to further differentiate?**
- **Hematopoietic growth factors= Colony Stimulating Factors (CSF):**
 - Cytokines stimulate proliferation of progenitor and precursor cells and promote cell differentiation + maturation within specific lineages
 - Cytokine= category of small protein's that are involved in cell signaling
- 3 types of growth factors: (all are cytokines)
 - **Colony stimulating factor (CSF)**
 - **Interleukin**
 - **Erythropoietin + thrombopoietin**

Cytokine	Major activity	Important sources
GM-CSF	Promotes CFU-GM mitosis and differentiation	T cells, endothelial cells
G-CSF	Promotes CFU-G mitosis and differentiation, facilitates neutrophil activity	Macrophages, endothelial cells
M-CSF	Promotes CFU-M mitosis and differentiation	Macrophages, endothelial cells
IL-1	Promotes proliferation of pluripotent hematopoietic stem cells	Macrophages and helper T-cells
IL-2	Stimulates activated T and B cell mitosis, induces differentiation of NK cells	helper T cells
IL-3	Promotes proliferation of pluripotent hematopoietic stem cells	Helper T cells
IL-4	Stimulates B and T cell activation and development of mast cells and basophils	Helper T cells
IL-5	Promotes CFU-Eo mitosis and activates eosinophils	Helper T cells
IL-6	Promotes proliferation of pluripotent hematopoietic stem cells	Macrophages, neutrophils, local endothelial cells
Erythropoietin (EPO)	CFU-E differentiation	Peritubular endothelial cells of kidney capillaries
Thrombopoietin (TPO)	CFU-Meg proliferation and differentiation	Kidney and Liver



MYELOBLAST - Earliest recognizable neutrophil precursor in bone marrow: Large, spherical nucleus. 3 to 5 nucleoli, small amount of **agranular** cytoplasm, stains basophilic (blue)

PROMYELOCYTES – Large spherical nucleus with azurophilic (**primary**) granules, **only made in promyelocytes**

MYELOCYTES - Nucleus becomes more heterochromatic, becomes **indented-Specific granules begin to form**

METAMYELOCYTES – Stage when neutrophil, eosinophil, and basophil can more definitely be identified based on presence of many specific granules- Nucleus indentation deepens becomes kidney-bean shaped

BAND (stab) -Nucleus elongated and horseshoe-like shaped, precedes development of first distinct nuclear lob
-When 2 to 4 nuclear lobes are seen, cell is mature neutrophil or **PMN, i.e., POLYMORPHONUCLEAR NEUTROPHIL**
aka segmented neutrophil

Hematopoiesis: Myelopoiesis

- Myeloblast, promyelocyte, **eosinophilic** myelocyte, eosinophilic metamyelocyte, **eosinophil**
- Myeloblast, promyelocyte, **basophilic** myelocyte, basophilic metamyelocyte, **basophil**
- **Cannot differentiate eosinophilic or basophilic precursors from neutrophilic precursors morphologically until late myelocyte stage or early metamyelocyte stage when specific granules appear**

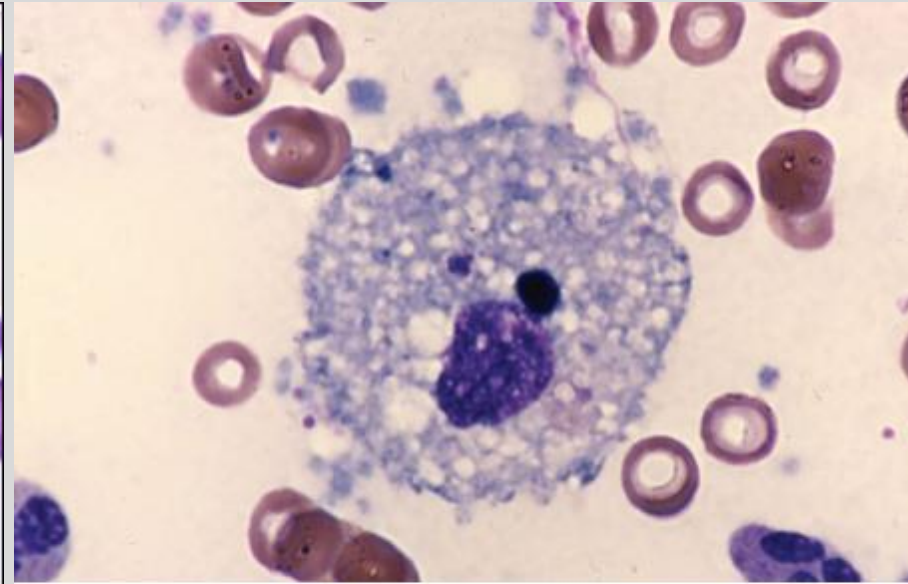
Hematopoiesis:

Myeloid lineage-Agranulocytes

Monocyte + Macrophage



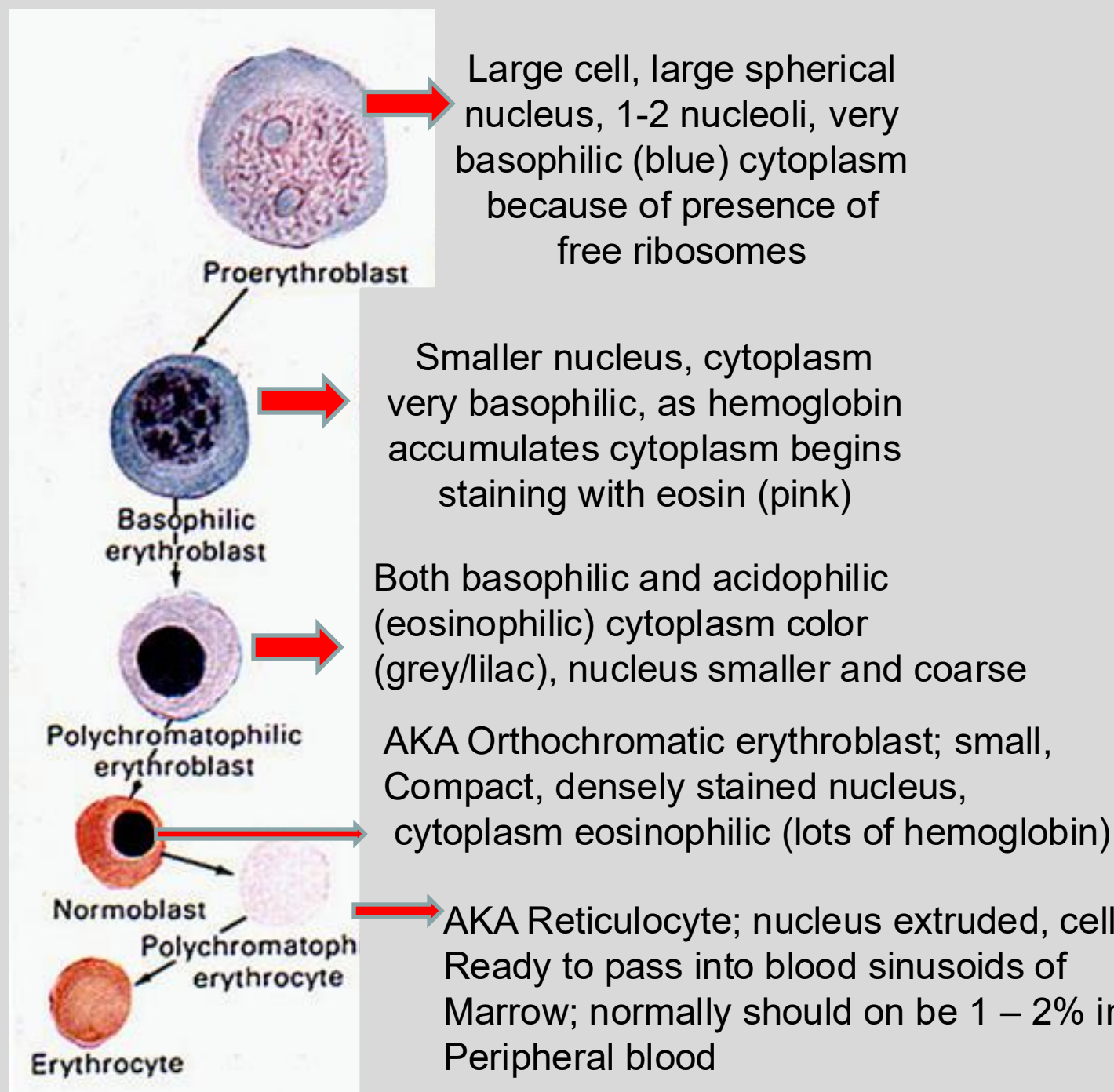
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Source: Lichtman MA, Kipps TJ, Seligsohn U, Kaushansky K, Prchal JT:
Williams Hematology, 8th Edition: <http://www.accessmedicine.com>
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Mature monocytes enter the bloodstream, circulate for a few hours, and then enter various tissues where they mature into **Macrophages** (or other phagocytic cells) and function for several months



Hematopoiesis: Thrombopoiesis

Thrombocyte= platelets:

- **Appearance:**
 - Small, anucleate cytoplasmic fragments derived **from megakaryocyte**
- **Action:**
 - Role in Primary hemostatic plug
 - Life-span= 10 days
- CFU-GEMM stem cell differentiates into CFU-Meg. Under the influences of the growth factors:
 - GM-CSF, IL-3, **thrombopoietin**

Megakaryoblast:
Non-lobed nucleus

Pro-megakaryocyte:

Megakaryocyte
Multilobulated nucleus,
azurophilic granules
Platelets derive from
this stage

Platelets
Cytoplasmic fragments
of megakaryocyte

Hematopoiesis – Thrombopoiesis

- Platelets produced in marrow from same multipotential stem cells (CFU-GEMM) as erythroid and myeloid series
- Under influence of GM-CSF and IL-3, stem cell differentiates into CFU-Meg, committed to development into megakaryoblast
- **Megakaryoblast large cell with nonlobed nucleus**
- Under stimulation by **thrombopoietin**, ploidy increases before chromosomal replications ceases - cell eventually becomes MEGAKARYOCYTE which then produces platelets (thrombocytes)
- Megakaryocyte has complex, multilobulated nucleus and scattered azurophilic granules. TEM: peripheral cytoplasm megakaryocyte appears divided into small compartments by invagination of plasma membrane – **platelet demarcation channels**

Clinical Case

- Which of the white cell lines would most likely be elevated in this **acute** infection of the gallbladder?
 - a) Lymphocytes
 - b) Neutrophils
 - c) Monocytes
 - d) Macrophages
- What does the elevated white blood count represent?
 - a) Infection
 - b) Inflammation
 - c) Stress
 - d) All of the Above

Clinical Case

- A 36-year-old man presented with a 6-day history of bleeding gums and abdominal pain in the left upper quadrant. He reported having had fevers, fatigue, decreased appetite, and unintentional weight loss of 10 lbs during the previous month.
- On physical examination,
 - red, swollen gingivae
 - gingival hyperplasia
 - tender submandibular lymph nodes
 - a palpable spleen
- Labs revealed
 - WBC 194,100 mm³ with 44% blasts
 - Platelets 12,000 mm³

Clinical Case

- Which cell line is responsible for the symptom of *bleeding gums*?
 - a) Monocytes
 - b) Platelets
 - c) B cells
 - d) Plasma Cell
- What hormone/molecule regulates platelet production?
 - a) Erythropoietin
 - b) Thrombopoietin
 - c) IL-1
 - d) IL-3
- With his WBC count (194,000 mm³ with 44% blasts), would you expect that he would be able to fight infections better?

Normal CBC values

- RBC count:
 - M: 4.5 - 5.9 million/mm³
 - F: 4.0 - 5.2 million/mm³
- Hematocrit:
 - M: 41 - 53%
 - F: 36 - 46%
- Hemoglobin:
 - M: 13.5 - 17.5 g/dL
 - F: 12.0 - 16.0 g/dL
- Mean Corpuscular Volume (MCV)
 - 80 - 100 μ m³
- Mean Corpuscular Hemoglobin (MCH)
 - 26.0 - 34.0 pg/cell
- Mean Corpuscular Hemoglobin Concentration (MCHC)
 - 31.0 - 37.0 g/dL
- Reticulocyte count:
 - 0.5 - 2.5%
- Platelet count:
 - 150,000 - 350,000/mm³
- Leukocyte count with differential
 - Leukocyte count:
 - 4,500 - 11,000/mm³
 - Segmented Neutrophils
 - 40 - 70%
 - Bands
 - 0 - 10%
 - Eosinophils
 - 0 - 8%
 - Basophils
 - 0 - 3%
 - Lymphocytes
 - 22 - 44%
 - Monocytes
 - 4 - 11%

Definitions

- Anemia:
 - any condition where the number of erythrocytes (RBCs), Hemoglobin, or Hematocrit are decreased
- Polycythemia:
 - increase in total number of RBCs in peripheral blood
- Thrombocytopenia
 - abnormally low platelets in peripheral blood
- Thrombocytosis
 - increase in the number of platelets in the peripheral blood
- Monocytosis
 - increased number of monocytes in peripheral blood
- Eosinophilia
 - increased number of eosinophils in peripheral blood
- Leukopenia
 - total number of leukocytes in peripheral blood less than normal
- Leukocytosis
 - total number of leukocytes in peripheral blood greater than normal
- Neutropenia
 - abnormally small number of neutrophils in the peripheral blood
- Neutrophilia
 - an increased number of neutrophils in the peripheral blood
- Lymphopenia
 - a reduction, relative or absolute, in number of lymphocytes in peripheral blood
- Lymphocytosis
 - an increase, actual or relative, number of lymphocytes in peripheral blood

Summary Slide

- Hematopoiesis – Fetus: yolk sac followed by hepatic phase; Infant all bone marrow; adult bone marrow mostly flat bones and proximal femur
- Red bone marrow – vascular, mainly newborns
- Yellow bone marrow – adipose tissue, adults
- Cellularity – normocellular, hypocellular, hypercellular, acellular, 100 – age +/- 10%
- Monophyletic theory of hematopoiesis - all blood cells arise from common stem cell, the pluripotential stem cell (CD34+), which gives rise to all other progenitor stem cells
- Bone marrow inability to keep up with demands results in extramedullary hematopoiesis in liver, spleen, lymph nodes

Summary Slide

- CFU – GEMM differentiates into CFU-GM, CFU-E, CFU-Meg, CFU-Eo, CFU-B; separate CFU-L (lymphocytes)
- Growth factors: CSFs (colony stimulating factor), IL (interleukin), erythropoietin, and thrombopoietin
- Review hierarchy of hematopoiesis for neutrophil, eosinophil, basophil, monocyte, red blood cell (erythrocyte), platelet (thrombocyte)
- Morphology and order of precursor cells – myeloblast>> promyelocyte>>myelocyte>>metamyelocyte>>band/stab>> PMN
- **Cannot differentiate eosinophil or basophil from neutrophil until late myelocyte/early metamyelocyte stage when specific granules appear**
- Morphology and order – proerythroblast>>basophilic erythroblast>>polychromatic erythroblast>> orthochromatic erythroblast>>reticulocyte>> erythrocyte (mature red blood cell)
- Thrombopoiesis: megakaryoblast>>Pro-megakaryocyte>> Megakaryocyte>>platelets